

3	(a) (i) $v^2 = 2as$ $1.2^2 = 2 \times a \times 1.9$ $a = 0.38 \text{ m s}^{-2}$	M1 A1 [2]
	(ii) $F = ma$ $= 42 \times 0.38$ $= 16 \text{ N}$	M1 A0 [1]
	(b) $\text{power} = Fv$ $= 16 \times 1.2$ $= 19 \text{ W}$	C1 A1 [2]
	(c) (i) $\text{component} = 42 \times 9.8 \times \sin 2.8$ $= 20.1 \text{ N}$	C1 A1 [2]
	(ii) $\text{accelerating force} = 20.1 - 16 = 4.1 \text{ N}$ $\text{acceleration of trolley} = 4.1 / 42 = 0.098 \text{ m s}^{-2}$ $s = \frac{1}{2}at^2$ $3.5 = \frac{1}{2} \times 0.098 \times t^2$ $t = 8.5 \text{ s}$	C1 C1 C1 A1 [4]

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- (d) *either* allows plenty of time to stop runaway trolley
or speed of trolley increases gradually
or trolley will travel faster B1 [1]
(answer must be unambiguous when read in conjunction with question)

- 1 (a) (i) 1 stress = force / (cross sectional) area B1 [1]